

*Research and Ethics
in Agricultural and Resource Economics:
Conducting Ethical Research*

Ethics 2

Year 1

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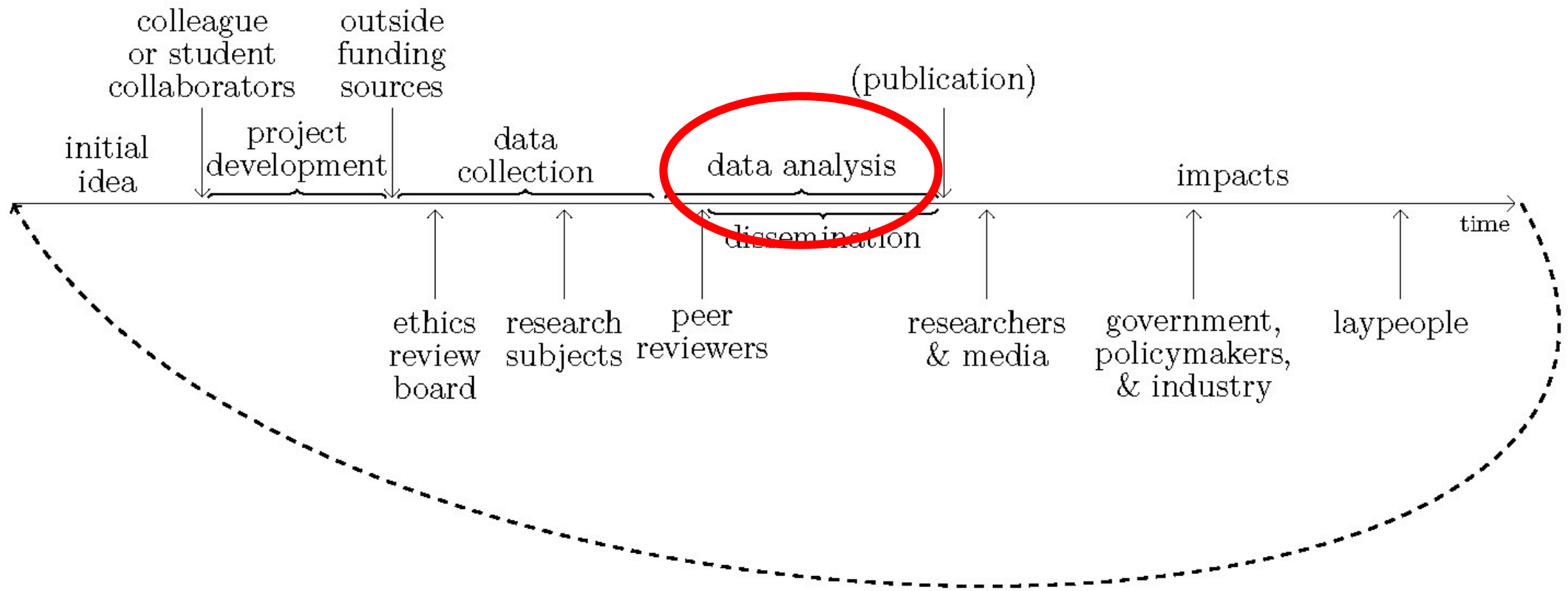
Adapted from Sarah Janzen (UIUC) & Jeff Michler (UA)

Outline

- Recall the problem
- Potential solutions
- Implementing best practices

Recall the problem

Dan Ariely, Michael LaCour, Brian Wansink



Data Manipulation: p -Hacking or HARKing

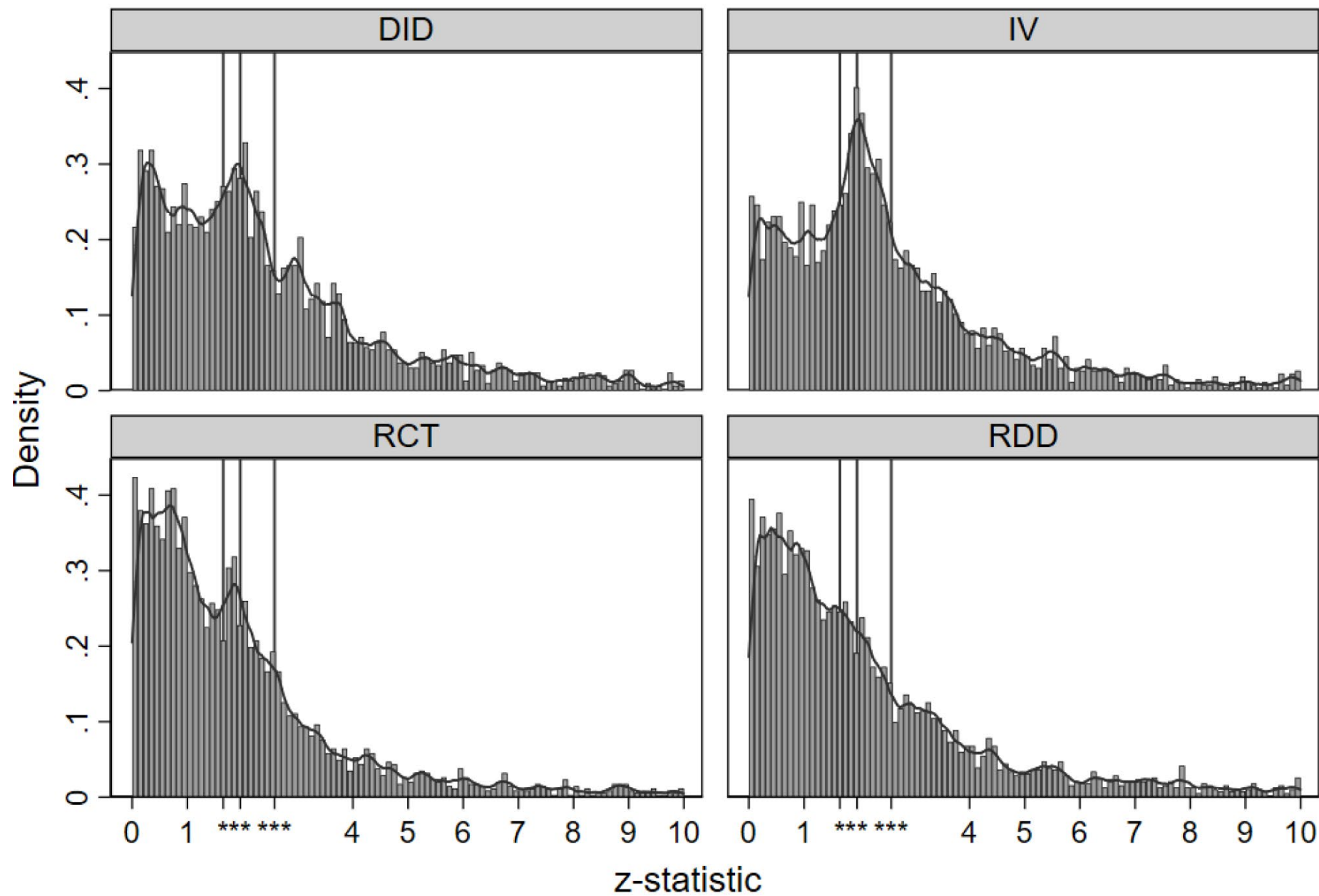
Manipulate the data or specification so that the **result you want** becomes significant.

1. *p -Hacking / fishing / data dredging / data mining / researcher degrees of freedom* – variable or sub-sample construction, specifications.
 - e.g., “Do not despair. It looks like stickers on fruit may work (with a bit more wizardry).”
2. *Hypothesizing After Results are Known (HARKing)* – running results to see what is significant, without a hypothesis as your guide.
 - Not guided by theory or conceptual model.

Defining and Describing Data Manipulation

- Leamer: “Let’s take the ‘con’ out of econometrics” (1983).
- Kennedy: “Sinning in the basement” (2002).
- Gelman and Loken: “Garden of forking paths” (2013):
“...dataset can be analyzed in so many different ways (with the choices being not just what statistical test to perform but also decisions on what data to [include] or exclude, what measures to study, what interactions to consider, etc.), that very little information is provided by the statement that a study came up with a $p < .05$ result.”

Brodeur et al. “Methods Matter” AER (2020)



Graphs by Method

Potential solutions

Pre-Analysis Plans, Transparency, Reproducibility

How Can We Change Incentives ?

- 1) Pre-commitments to “tie hands”
 - 1) Pre-analysis plans
 - 2) Journals accept research plans then publish results
 - 3) Hypothesis registry
- 2) Transparency and Reproducibility
 - 1) Lab notebook approach & submission requirements
 - 2) Transparency when search over many models performed
 - 3) Reproducible workflow

What Is A Pre-Analysis Plan ?



They bound me hand and foot in the tight ship—
erect at the mast-block, lashed by ropes to the mast.

— Homer, The Odyssey, Book XII

A **pre-analysis plan** is a step-by-step plan setting out how a researcher will analyze data which is written in advance of them seeing this data (and ideally before collecting it in cases where the researcher is collecting the data).



What Can A Pre-Analysis Plan Accomplish ?

1. Restrict **which hypotheses** to investigate.
2. Restrict **how to test** the hypotheses.
3. Pre-commit to a **data collection** plan.

The result is that a pre-analysis plan should increase the probability that a published positive result is true.

(Coffman and Neiderle, 2015, JEP)

Pre-Analysis Plans in Medicine

- Plans has been required by law in the United States for all drug trials since 1997
 - Stated goal was to counter strong incentives of drug companies to find positive results
- In 2004 the twelve journals of International Committee of Medical Journal Editors announced public trial registration would be pre-requisite for publication
 - Stated goal was to counter the “file drawer problem,” where only significant results get published
- ClinicalTrials.gov is where most are registered
 - Pre-2004, registry averaged 1,879 a year
 - Post-2004, registry averaged 22,088 a year

Pre-Analysis Plans in Social Science

- Pre-analysis plans predate registry
- Most commonly cited first studies:
 - Casey, Glennerster, and Miguel (2012). “Reshaping institutions: Evidence on aid impacts using a preanalysis plan.” QJE.
 - Finkelstein et al. (2012). “The Oregon health insurance experiment: Evidence from the first year.” QJE.
- In 2012 AEA established the AEA RCT Registry
 - Allows for registration of lab and field experiments
- Additional social science registries founded since
 - 3ie’s Registry for International Development Impact Evaluations - impact evaluations in low income countries
 - Open Science Foundation registry - any study

Pre-Analysis Plans v. Registrations

- In medical science, there has been a clear distinction between:
 - What pre-analysis plans accomplish - p-hacking
 - What public registration accomplishes - file drawer problem
- In economics, these two have been muddled
 - See Olken (2015); Coffman et al. (2015); Burlig (2018); Duflo et al. (2020)
- Making pre-analysis plans work means understanding the difference between what a plan accomplishes and what a public registry accomplishes



Basics of a Pre-Analysis Plan

- Data description
 - Describe the process by which data was generated or collected
- Hypotheses
 - Clearly state the research questions and hypotheses to be tested
- Outcome variables
 - List the primary outcome variables as well as any secondary outcomes to be tested
- Variable construction
 - Precisely define how each outcome and control variable is to be constructed
- Model specification
 - Describe the statistical method of analysis, including covariates and the approach to inference



Comprehensive Pre-Analysis Plan Adds:

- Motivation
 - Provide motivation and context for the study. A review of the literature can be helpful
- Theoretical model or theory of change
 - The model to be tested or causal diagram that justifies the selected hypotheses
- Multiple inference adjustment
 - Address how multiple inference will be dealt with. Either FWER or FDR
- Data cleaning
 - Outline basic operating principles for how data cleaning will be conducted, such as windsorizing outliers, correcting for attrition, and adjusting for contamination
- Subgroup analysis
 - Describe any subgroup or heterogeneity analysis (sex, age, education, race)
- Power calculations
 - If data allows, include information on sample size and power calculations

Prevent p -Hacking

- Forming a Ulysses Pact is a good way to prevent p -hacking
- They bind the researcher to limit research degree of freedoms
- Important when there are strong incentives to cheat (drug trials)
- Or when there is leeway for what researchers can “discover”
 - Coffman and Niederle (2015) - economic theory limits what a research might “find”
 - Brodeur et al (2016, 2020) - little evidence of p -hacking in experiments

Solve the File Drawer Problem

- Ulysses Pacts don't solve the file drawer problem, but public registries do
- Regardless of the empirical methods a “hypotheses registry” creates database on the universe of research questions asked
- Pre-analysis plans and public registries work together but are not the same thing
 - Coffman and Niederle (2015) - promote basic plan and public registration
 - Registered reports at *Journal of Development Economics*, the *Journal of Accounting Research*, *The Review of Financial Studies*, and *PLOS One*

Ulysses' Pact or Ulysses' Raft

- Pre-analysis plans are a Ulysses' pact
 - They bind the researcher to a plan
 - They prevent p-hacking but also the ability to explore
 - Registering them helps solve the file drawer problem
 - In econ, focus has been on RCTs
- Typical econ uses a Ulysses' raft
 - Allows researchers to explore messy social science data
 - Does not limit the questions we can ask to what we think we might have known ahead of time
 - Could register exploratory work to solve the file drawer problem
 - In econ, been excluded from pre-analysis and registry discussion

Ulysses' Packraft

Of immense value in a country where the want of wood renders it impossible to form any sort of raft.
— French explorer Joseph Renè Bellot



- A packraft is a portable inflatable boat first designed in the 1840s for use in exploring the Canadian arctic
- The researcher develops a basic pre-analysis plan and files it with a public registry
- Unlike a binding pact, researchers are allowed to update the plan as new information arises as in arXiv.org
- Transparency about deviations from the original plan allows for flexibility, while also addressing issues of p -hacking
- Encourages a disciplinary practice of telling readers at what point in the research process (and for what reason) a particular model emerged

Implementing best practices

Code, Workflow, Version Control

Changes in Culture

- Institutional responses
 - Open Science Foundation
 - Berkeley Institute for Transparency in the Social Sciences
- Journal policies
 - Transparency and Openness Promotion
 - “The *AJAE* adheres to TOP “Level 1” standards as described here...”

Data and Documentation

Data sources should be indicated clearly in the article. Upon acceptance and in keeping with evolving policies at other top economics journals, authors are expected to submit their datasets and associated documentation (perhaps in a readme file), for readers to download from the AJAE website. Authors are also expected to provide whatever other material is needed to ensure that their results can be replicated (this might include code or pseudo-code used in estimation). Authors are encouraged to comply with all of this policy, but the editors would prefer partial compliance over non-compliance. The submitted files will be accessible as clickable links to the online version of the published paper. It is understood that in cases where the data are proprietary or confidential authors cannot fully satisfy this policy. Such cases should be brought to the attention of the editor at the time of initial submission.

Transparency and Openness

- Code and data management have improved with this openness expectation – less data sloppiness
 - Data notebook
 - JPAL/CEGA offer members services of their replication team: barriers to entry

```
131 drop _merge
132 save "C:\Users\aljosephson\Dropbox\Out for Review\DISE2_Sorghum\newrebuild_10Jan-withstaple_rain.dta", replace
133 probit panelhh grow_sorghum tot_sorghum_area grow_maize tot_maize_area plot_area assetvalue head_age head_edu num_cattle plough rec_ext sorc
134 tab plot_area, missing
135 tab head_age, missing
136 sort yearpanel
137 sort panelhh
138 by panelhh: tab head_age
139 by panelhh: tab head_age, missing
140 *tried to come up with a better method ... but didn't find one
141 *so by hand replaced information, based on, if household head was the same, and thus could determine if someone aged, by how much, etc.
142 *still missing about 100 values for head_age and head_edu
143 sort rc id yearpanel head_age
144 order rc id yearpanel head_age head_edu
145 replace head_age = 71 in 8
146 replace head_age = 60 in 11
147 replace head_age = 54 in 13
148 replace head_edu = 5 in 13
149 replace head_age = 43 in 26
150 replace head_edu = 11 in 26
151 replace head_age = 56 in 33
152 replace head_edu = 8 in 33
```

- Will enhanced transparency provide sufficient information to detect misconduct?

Version Control with GitHub

Current repository

covid_food_security

Current branch

fies

#1

Fetch origin

Last fetched 2 minutes ago

Changes

History

Select branch to compare...

changed edit date to match

44iggs44 • 56m

deleted files, cleaned files, new pathways

44iggs44 • 1h

changes to saving in eth_build

jdmichler • 2h

integrate wave_00 fies into eth panel

jdmichler • 2h

updated baseline fies cleaning

jdmichler • 2h

finished cleaning bf_fies_0

44iggs44 • 1d

added wave variables to nigeria, create...

44iggs44 • 1d

Got rid of unnecessary file, added var t...

44iggs44 • Sep 14, 2021

final edits changed fies_2

44iggs44 • Sep 14, 2021

Split nga_fies_0 into pp and ph

44iggs44 • Sep 13, 2021

merged HH data into eth_fies_0

44iggs44 • Sep 10, 2021

Verify Code is Clean

44iggs44 • Sep 9, 2021

Update README.md

Jeffrey D. Michler • Sep 9, 2021

update project directories

jdmichler • Sep 9, 2021

Merge branch 'main' of https://github....

jdmichler • Sep 9, 2021

update project name

jdmichler • Sep 9, 2021

Update README.md

Jeffrey D. Michler • Sep 9, 2021

finished cleaning bf_fies_0

44iggs44

d3edda1

3 changed files

+24 -49

New

burkina_faso\bf_fies_0.do

nigeria\nga_fies_ph0.do

nigeria\nga_fies_pp0.do

33 +

34 34 *****

35 35 **# 1 - FIES data

36 36 *****

37 37

38 38 * load data

39 - use "\$root/menage/s08a_me_BFA2018", clear

39 + use "\$root/wave_00/menage/s08a_me_BFA2018", clear

40 40

41 41 * replace counts with binary indicators

42 42 lab def yesno 1 "Yes" 0 "No"

43 - foreach x in a b c d e f g h{

44 - replace s8q02`x' = 1 if s8q02`x' > 1 & s8q02`x' < .

45 - lab val s8q02`x' yesno

43 + foreach x in 1 2 3 4 5 6 7 8{

44 + replace s08aq0`x' = 0 if s08aq0`x' > 1 & s08aq0`x' < .

45 + lab val s08aq0`x' yesno

46 46 }

47 47

48 - replace s8q01 = 0 if s8q01 == 2

49 - lab val s8q01 yesno

50 -

51 -

52 48 * rename variables

53 - rename s8q01 fies_4

54 - rename s8q02a fies_5

55 - rename s8q02b fies_6

56 - rename s8q02c fies_8

57 - rename s8q02d fies_7

58 - rename s8q02e fies_2

59 - rename s8q02f fies_9

60 - rename s8q02g fies_1

61 - rename s8q02h fies_3

49 + rename s08aq01 fies_4

50 + rename s08aq02 fies_5

51 + rename s08aq03 fies_6

52 + rename s08aq04 fies_7

53 + rename s08aq05 fies_8

54 + rename s08aq06 fies_1

55 + rename s08aq07 fies_2

56 + rename s08aq08 fies_3

62 57

Bringing “Sinning” Out of the Basement

- “Sinners are not expected to avoid sins; they need only confess their errors openly.”
 - Leamer (1978)
- The goal should be open and transparent analysis
- Pre-define a research plan, even if you don’t develop a full PAP
- Keep meticulous records
- Ask questions
- Learn by doing

Homecoming

